

# The development of laser-produced plasma EUV light source



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Cite as: Yang, D.-K. et al. The development of laser-produced plasma EUV light source. *Chip* **1**, 100019 (2022).  
<https://doi.org/10.1016/j.chip.2022.100019>

Received: 31 May 2022

Accepted: 6 July 2022

Published online: 13 July 2022

Extreme ultraviolet lithography (EUVL) has been demonstrated to meet the industrial requirements of new-generation semiconductor fabrication. The development of high-power EUV sources is a long-term critical challenge to the implementation of EUVL in high-volume manufacturing (HVM), together with other technologies such as photoresist and mask. Historically, both theoretical studies and experiments have clearly indicated that the CO<sub>2</sub> laser-produced plasma (LPP) system is a promising solution for EUVL source, able to realize high conversion efficiency (CE) and output power. Currently, ASML's NXE:3400B EUV scanner configuring CO<sub>2</sub> LPP source system has been installed and operated at chipmaker customers. Meanwhile, other research teams have made different progresses in the development of LPP EUV sources. However, in their technologies, some critical areas need to be further improved to meet the requirements of 5 nm node and below. Critically needed improvements include higher laser power, stable droplet generation system and longer collector lifetime. In this paper, we describe the performance characteristics of the laser system, droplet generator and mirror collector for different EUV sources, and also the new development results.

**Keyword:** Extreme ultraviolet source, Laser-produced plasma, Sn target, Long lifetime collector

## INTRODUCTION

The development and growth of semiconductor industry have always followed Moore's law in the recent 50 years, and it is fair to say that the

continuous shrinking of transistor dimension in microchips was driven by the development of photolithography technology<sup>1–3</sup>. The first G-line (436 nm wavelength) step-and-repeat photolithography system appeared in late 1970s, and it dramatically reduced the feature size of the devices on a chip<sup>4</sup>. According to the Rayleigh diffraction limit of resolution, to get a smaller feature in photolithography, shortening the exposure wavelength ( $\lambda$ ) is the most readily accessible solution, compared with reducing the production process parameter  $k$  and increasing the numerical aperture (NA)<sup>5–7</sup>. In this situation, light sources continue changing, from high-pressure mercury arc lamp for g/h/i-line (365 nm)<sup>8–13</sup>, to krypton fluoride (KrF) excimer laser for 248 nm deep ultraviolet<sup>14–18</sup>, and to argon fluoride (ArF) for 193 nm deep ultraviolet (DUV)<sup>19–23</sup>. The implementation of immersion ArF scanner<sup>24–28</sup>, combined with the resolution enhancement technologies (RET) and multi-patterning technologies, enabled the semiconductor industry to realize 7 nm node<sup>29</sup>. However, to further shrink the feature size to 5 nm and below, more process steps and masks are needed in a single functional layer for the multi-patterning. This means an increase in material cost, equipment cost and process complexity, while yield control becomes challenging.

To reduce the process complexity of DUV lithography, extreme ultraviolet lithography (EUVL) was proposed, which used 13.5 nm wavelength light source<sup>8</sup>. It has been considered as a leading-edge technology for the fabrication of new-generation semiconductor devices at the 5 nm node and below. The main challenge of EUVL is to achieve high enough wafer throughput in high-volume manufacturing (HVM). The difficulty is the high absorption of the extreme ultraviolet (EUV) light by any inter-medium. To resolve this problem, vacuum environment is implemented and the lens system is replaced by a mirror system for imaging. Even with such measures, the power loss is still large. For a mirror with Mo/Si multi-layer reflection coating, the maximum reflectivity of EUV can only reach about 75% theoretically<sup>30</sup>. Such a high loss brings high requirements to the power of the EUV light source. In order to achieve a throughput of 100 wafers per hour (WPH) at a typical scanner duty cycle of 60%, the power at the wafer surface should be greater than 550 mW<sup>31</sup>. However, there is high power loss, which is mainly due to the reflectivity loss of 12 mirrors and mask. In addition, the limited transmittance of the pellicle located on top of the masks will further increase the loss. This means that EUV source power provided to the exposure tool at intermediate focus have to be greater than 200 W<sup>32</sup>. As such, an efficient, clean, and high-power EUV light source need to be developed.

In previous works, four basic technologies have been developed to generate EUV light sources:

- (1) Synchrotron radiation facility (SRF)<sup>33–36</sup>;
- (2) Discharge-produced plasma (DPP)<sup>37–43</sup>;

- (3) Free-electron laser (FEL)<sup>44–46</sup>; and
- (4) Laser-produced plasma (LPP)<sup>47–62</sup>.

In the course of research of laser-produced plasma, the most striking discovery is that, with the atomic number ( $Z$ ) of the target materials increasing, the excited spectral wavelengths become shorter. In an early study, LPP emission spectra were observed for targets in the range  $Z = 55 - 71$ . With increasing  $Z$ , the emission peak wavelength moved from 20 nm to 4 nm<sup>63</sup>. However, the emission features in the short wavelength range became more complicated due to the 4d–4f transition of several adjacent stages. Therefore, emission of strong isolated lines is inhibited, resulting in no apparent peak wavelength. More materials with different  $Z$  have been investigated in recent research<sup>64</sup>. Sn has been considered the most hopeful target due to that its excited spectra are dominated in the 13.5 nm region<sup>65,66</sup>. The spectra mainly arose from  $4p^64d^n - 4p^54d^{n+1} + 4d^{n-1}4f$  transitions. This performance was attributed to the fact that there is significant overlap between the lines of these transitions at different ion stages.

In a free-electron laser (FEL), a beam of electrons travel through a transverse periodic magnetic field to amplify electromagnetic radiation<sup>67</sup>. Similar to conventional lasers, FEL can be configured as an amplifier or an oscillator<sup>68</sup>. In principle, the resonant output wavelength of FEL is fully tunable, and is given by  $\lambda_R \propto (1 + K^2)/\gamma^2$ , where  $K$  is the normalized undulator strength and  $\gamma$  is the electron beam's relativistic Lorentz factor<sup>69</sup>. This indicates that shorter  $\lambda$  has lower peak power levels, resulting from normalized undulator strength. In this situation, high-gain harmonic generation technologies are proposed as a promising approach<sup>70–72</sup>. In this approach, the coherence properties of a (relatively) long-wavelength laser are transferred to a shorter-wavelength FEL output. To date, several EUV and x-ray FEL facilities have been constructed around the world and applied to a variety of cutting-edge researches<sup>68,69,73,74</sup>. Despite that these high-gain FELs meet most criteria for light source applications, further technological advancements are needed to realize 13.5 nm EUVL. Examples of such improvements include, improved relatively low repetition rate, significant output pulse energy jitter and limited number of user beamlines. To further promote the application of FEL in EUVL, several improved methods are proposed<sup>75,76</sup>. However, these schemes are still at the theoretical stage and no practical applications have so far been published.

Another approach to EUV source is SRF. Synchrotron radiation is emitted when charged particles are accelerated in a magnetic field at almost the speed of light. Three types of magnetic structures are generally used to generate synchrotron radiation: bending magnets, undulators, and wigglers<sup>77</sup>. Bending magnets force electrons to move along a curved path, meanwhile, a fan radiation is emitted with a broad spectral range from microwave to X-ray. In this case, the emission angle is  $1/\gamma$ , where  $\gamma$  is the Lorentz contraction factor<sup>78</sup>. Undulator radiation is generated when the electrons pass through a relatively weak periodic magnetic field. This weak magnetic field causes a small oscillation amplitude of the undulation, and leads to radiation with small emission angle  $1/\gamma\sqrt{N}$  and narrow spectrum, where  $N$  is the number of magnet periods. Same as undulator radiation, wiggler radiation is also produced in a periodic magnetic field. The difference is that the periodic magnetic field used in wiggler radiation is very strong. Due to the strong magnetic field, the oscillation amplitude is large. It causes the wiggler emission angle  $K/\gamma$  to be bigger than those of undulator radiation and bending magnet radiation, where  $K$  is the electron deflection parameter.

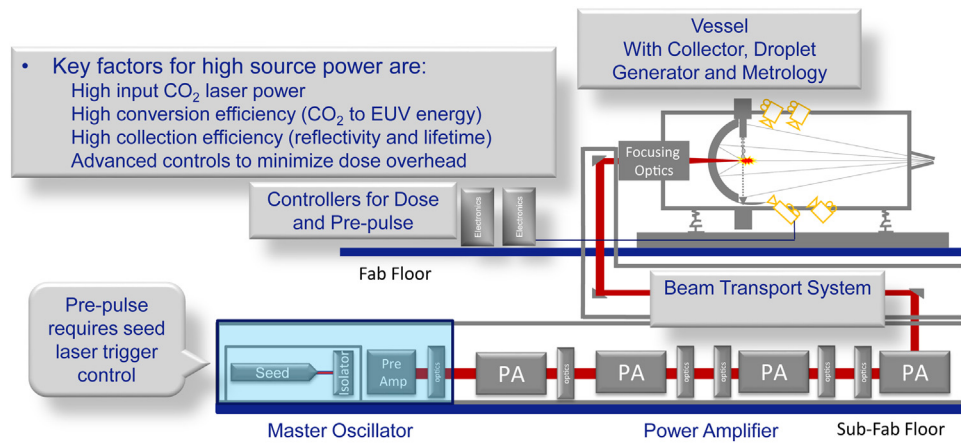
EUV imaging experiments using synchrotron radiation as source has been studied for a long time<sup>79–81</sup>. Shanghai Synchrotron Radiation Facility (SSRF), as one of third-generation SRF, has storage rings composed of many long straight sections for undulators and wigglers<sup>82</sup>. SSRF has 15 beamlines in the first phase of the project, and 26 beamlines after the

second phase is completed. It provides facilities for the research fields of life sciences, chemical science, physics and materials science, medicine, earth and environmental sciences, interdisciplinary, archaeology, and industrial applications. Currently, the soft X-ray interference lithography (XIL) beamline of SSRF is used to produce EUV light, which can manufacture 1D and 2D periodic nanostructures<sup>33</sup>. The results demonstrated that XIL beamline has a high performance, with 50 nm half-pitch 1D and 2D patterns from a single exposure area of  $400\ \mu\text{m} \times 400\ \mu\text{m}$ . However, the SRF is not suitable for HVM due to its complex device structure, enormous size and high cost.

In the case of a DPP source, an electrical discharge is struck in the target material, producing a plasma that emits radiation with EUV wavelength. A number of DPP projects have been built up or are currently being built as potential solutions for EUVL. To meet the requirements of EUVL, the main criteria for working material selection are high conversion efficiency (CE) and low produced debris. Xe was the first emitting material candidate for DPP source due to its cleanliness. But Xe-based DPP technology has a rather low CE (0.5–1.2%), because the radiation peak at 13.5 nm is not intensive, leading to large energy loss<sup>3,37</sup>. To increase CE, a straightforward solution is to choose a working material with a very intensive radiation peak at 13.5 nm. Therefore, the Sn-based DPP configuration was investigated due to its intensive radiation peak at 13.5 nm. Using Sn-based DPP technology with fixed electrodes, a higher CE range of 1.4%–2% can be achieved compared with Xe-based DPP technology<sup>37</sup>. However, erosion in fixed electrodes will occur, caused by large heat load from the discharge plasma<sup>83</sup>. The corroded electrode materials are introduced onto the collector as debris contamination, which reduces the lifetime of the collector mirror. In order to reduce the electrode erosion, a new type of Sn-based DPP source consisting of two rotating electrodes was employed<sup>39,40,43</sup>. Generally, the debris generated by Sn-based DPP sources can be divided into three different types in addition to electrode erosion: the micro-particles or liquid Sn droplets, the slow atomic-ionic debris and the fast ionic debris<sup>83</sup>. Furthermore, by using moving electrodes, it is possible to modify the angular distribution of the micro-particles: particles get additional momentum compared with fixed-electrode DPP, which allowed the decrease of Sn droplet debris. Sn droplet debris can be further decreased by increasing the electrode velocity<sup>41</sup>. In order to obtain higher EUV power, Philips has proposed an innovative Sn DPP source system based on rotating electrode and an assisted configuration<sup>39</sup>. As a result, a CE of about 2% at 200 W deliver EUV power in  $2\pi$  steradian (sr) has already been demonstrated. However, the problem of optics protection against debris consisting of slow atomic-ionic and fast ionic debris for all kinds of DPP configurations is still difficult. To overcome the difficulties associated with the debris of DPP and the drawbacks of FEL and SRF, LPP has become the most promising technology for the radiation of EUV light source.

In the case of LPP, the target material is evaporated, heated and ionized by the high-power laser pulse to emit the EUV wavelength radiation in plasmas. In the early development of LPP EUV sources, the first consideration is minimizing the mass of target material to reduce unnecessary generation of debris. In order to limit the size of the target interacting with the high-power laser, several designs have been investigated and tested over the years. Starting from gaseous<sup>84–86</sup>, liquid<sup>87</sup>, and solid Xe<sup>51</sup>, to Sn droplet<sup>47,48,78–80</sup>, these materials have been used as targets for plasma production. As mentioned in the DPP section, Xe has a rather low CE due to the radiation peak at 13.5 nm being not intensive. For this reason, the current LPP target material mostly focuses on Sn droplet.

ASML subsidiary Cymer and Japan-based company Gigaphoton are long-term competitors, as both are trying to develop high-power and high-CE Sn LPP light sources for EUVL. The typical schematic diagram of LPP source is shown in Fig. 1<sup>31,88</sup>. It consists of five main components:



**Fig. 1 | Schematic of LPP EUV source system.** Reprinted with permission from ref.<sup>88</sup>. © 2017 Source Workshop.

(a) the high-power CO<sub>2</sub> laser, comprising of a master-oscillator power-amplifier (MOPA), (b) a beam transport system, including focusing and beam position control, (c) a vacuum vessel to generate Sn droplet, (d) a collector mirror to focus the EUV radiation, and (e) metrology modules. The laser-plasma interaction will take place when the CO<sub>2</sub> laser light is focused onto a Sn droplet generated by the vacuum vessel. Then the plasma distribution is measured with optical metrology modules, providing feedback to keep synchronization and optimal performance. Lastly, EUV light will be collected by an ellipsoidal mirror with  $2\pi$  sr.

## LASER SYSTEM OF Sn LPP

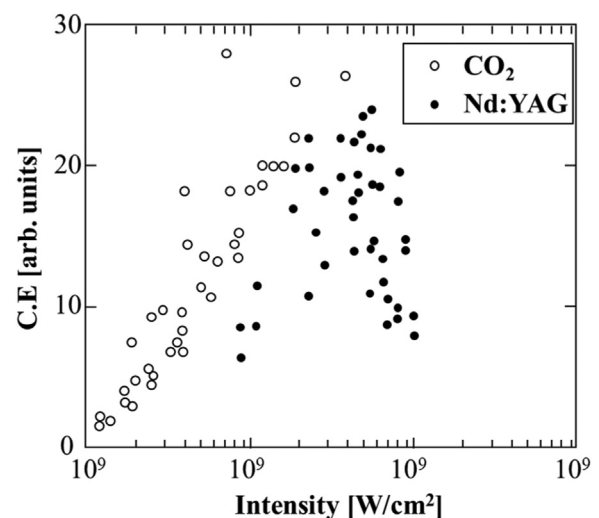
Plenty of research has been done on EUV light source with Sn LPP configurations. Most of them fall into one of the two categories with respect to the excitation lasers: CO<sub>2</sub> gas lasers at 10.6  $\mu\text{m}$  wavelength, and Nd:YAG solid-state lasers at 1064 nm wavelength. The technique of using Nd:YAG lasers to excite Sn droplet and produce plasma has been exclusively investigated. The CE from the laser to the EUV light lies in the range from 0.5% to 2% per  $2\pi$  sr at 13.5 nm (2% B.W.).<sup>51,62,84–87,89</sup> Since the CE of the Nd:YAG laser system is low, it is necessary to further increase the power to meet the requirements of HVM. Achieving the required  $> 550$  mW exposure power posts a significant challenge. Both theory and experiment<sup>90</sup> have clearly indicated that, Sn LPP based on CO<sub>2</sub> laser system has higher potential than that based on Nd:YAG laser system in realizing high CE and high power. As shown in Fig. 2, the CE of a Sn LPP based on Nd:YAG laser system reaches its maximum at a laser intensity of about  $5 \times 10^{10} \text{ W cm}^{-2}$ , and decreases with higher laser intensity. As a comparison, the CE monotonically increases with the increasing of laser intensity, in the case of the Sn LPP based on CO<sub>2</sub> laser system. However, the curve stopped at the laser intensity of  $2 \times 10^{10} \text{ W cm}^{-2}$ , due to the limitation on the focusing ability of CO<sub>2</sub> laser. Further CE improvement requires the increasing of CO<sub>2</sub> laser power, and laser power scaling was also proposed as a potential solution. With the development of laser power scaling in CO<sub>2</sub> laser, Sn LPP based on CO<sub>2</sub> laser system has proven to be the most promising solution to generate high-power EUV source<sup>91</sup>.

EUV power scaling has advanced significantly, thanks to the modification of the laser architecture. Historically, three-drive laser architectures were employed, and the corresponding EUV powers are shown in Fig. 3<sup>92</sup>. These architectures include no master oscillator (NOMO), MOPA, and MOPA with pre-pulse (PP)<sup>93</sup>. NOMO technique generates plasma by placing a grating or a mirror on the backside of the three amplifiers. When the tin droplet reaches the focal region of final focus element, it acts as a

mirror to establish a laser cavity, as shown in Fig. 4. If the gains exceed losses in the cavity, a relatively high-power laser beam is generated and delivered to the droplet for EUV radiation. This technique is applied to ASML's NXE:3100 light sources due to its simplicity and laser-to-droplet spontaneity. However, this configuration only produced 8 kW laser power and 10 W EUV power, with an average CE of around 0.8%<sup>94,95</sup>.

In order to produce higher EUV power via laser power scaling, MOPA and MOPA + PP techniques (Fig. 5) were proposed<sup>93</sup>. Both techniques include a master oscillator, which generates a laser pulse to seed the power amplifiers for single-pass gain extraction. The seed pulse is triggered by the tin droplet, so the laser pulse and the tin droplet arrive at the primary focus at the same time. The MOPA technique was first combined with the NXE:3100 three-stage power amplifier system for research throughout 2011 and 2012. An EUV power of 30 W was collected using this configuration. Since pre-pulse technology was not used, the architecture obtained relatively low CE of around 1%.

After adding the PP technology to the MOPA architecture, the CE was increased significantly. The schematic diagram of the PP technology is shown in Fig. 6<sup>88</sup>. Two laser pulses are used: PP illuminates the Sn



**Fig. 2 | Conversion efficiency (CE) from laser light to 13.5 nm light as a function of laser input intensity with a solid tin target.** Reprinted with permission from ref.<sup>90</sup>. © 2005 AIP Publishing LLC.

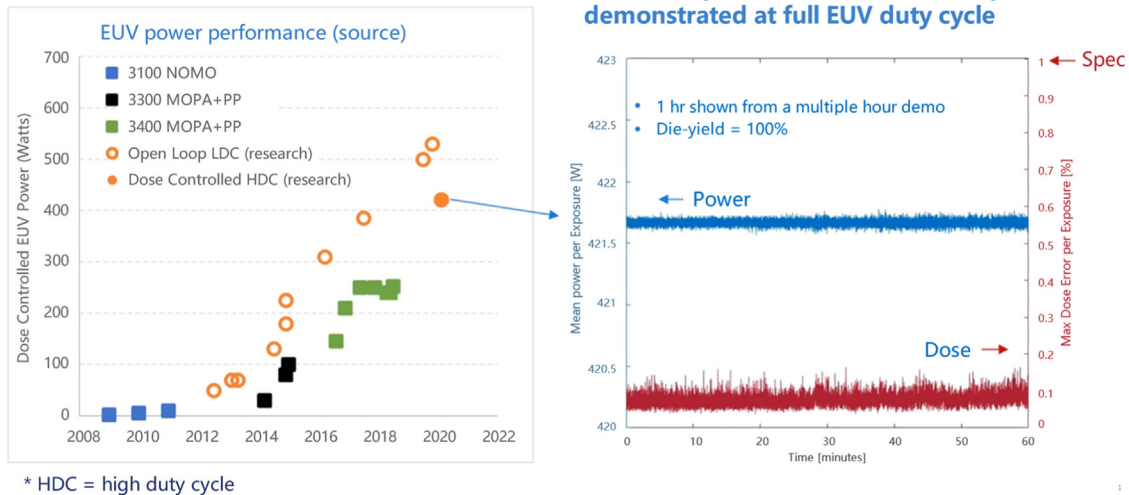


Fig. 3 | Historical EUV power scaling trend using LPP sources. Reprinted with permission from ref.<sup>92</sup>. ©2021 SPIE.

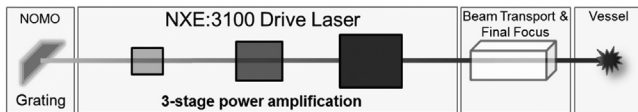


Fig. 4 | Schematics of the NOMO technique used in the NXE:3100 EUV light source. Reprinted with permission from ref.<sup>93</sup>. ©2015 SPIE.

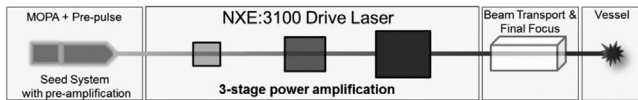


Fig. 5 | Schematics of the NXE:3100 MOPA pre-pulse EUV light source. Reprinted with permission from ref.<sup>93</sup>. ©2015 SPIE.

droplet to produce an expanded Sn cloud target, then another main pulse focuses onto the Sn cloud target to generate the required 13.5 nm plasma as the EUV light source. Fig. 7 illustrates the relationship between CE and target size as formed by PP<sup>93</sup>. Indeed, the MOPA + PP technique

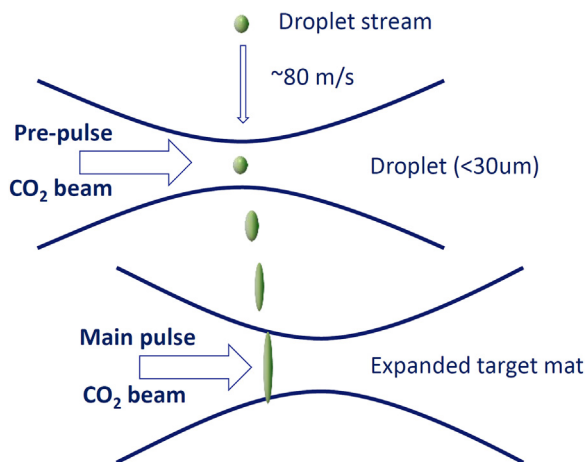


Fig. 6 | Spatial view of the target formation and EUV generation process. Reprinted from ref.<sup>88</sup>. © 2017 Source Workshop.

delivered around 4% CE, showing a significant improvement over MOPA. Such result demonstrates that PP is a key EUV power scaling method.

In 2014, the next-generation NXE:3300 system produced > 40 W of dose-controlled EUV power. Immediately afterwards, the laser power was increased to 80 W, and then to 100 W. In order to scale EUV power beyond 100 W and provide greater pre-amplification, ASML started to investigate high-power seed system (HPSS). Meanwhile, the first high-power drive laser (HPDL) was initiated with four-stage power amplifiers. In 2015, the configuration integrated HPSS with HPDL based on a standard NXE:3300, producing laser pulses which had roughly 2–3 times higher peak power than the standard NXE:3300. Furthermore, the average CE obtained through this configuration exceeds 5%. However, the new laser system with higher peak power leads to unstable laser onset, further reducing laser gain. In order to obtain a more stable laser, the isolation between amplifiers needs to be enhanced. The result has been demonstrated, as the laser gain of NXE:3400 is more than twice of NXE:3300<sup>88</sup>. After modification, the NXE:3400B scanner achieved 250 W/125 WPH with a

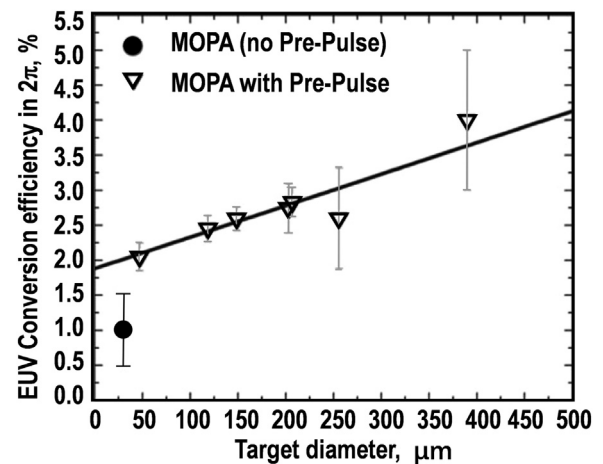
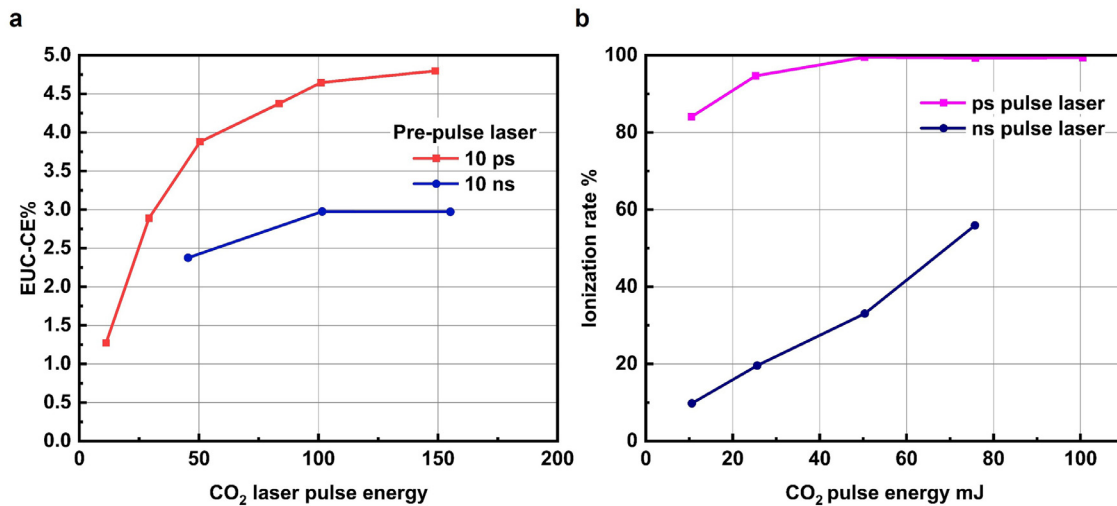


Fig. 7 | Conversion efficiency as a function of target diameter formed by the pre-pulse. All data were collected on an NXE:3100 MOPA pre-pulse stand-alone source. Vertical bars show standard deviation of 1,500 sequential pulses. Reprinted with permission from ref.<sup>93</sup>. © 2021 SPIE.





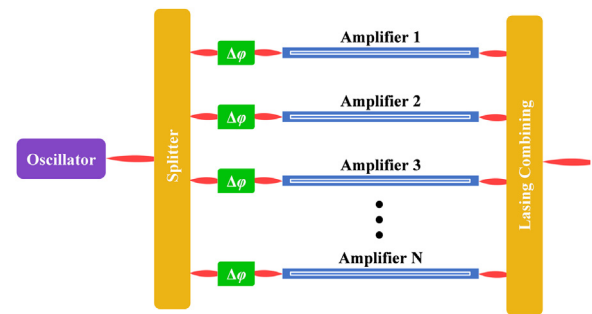
**Fig. 8 | CE and ionization under different PP laser conditions. a,** CE vs. CO<sub>2</sub> laser energy. **b,** Ionization ratio vs. CO<sub>2</sub> laser energy. Reprinted with permission from ref.<sup>99</sup>. © 2016 SPIE.

CE of 6%, and was released to semiconductor fabs in 2017<sup>96</sup>. This performance is further improved in the next-generation NXE:3400C, achieving 250 W/170 WPH at 30 mJ cm<sup>-2</sup> exposure dose<sup>97</sup>. Currently, study of the next laser power scaling is in progress by the research team at ASML. In a latest report, the dose-controlled EUV power reached 420 W at high duty cycle, and an open-loop EUV power of 530 W was obtained based on NXE:3600D<sup>98</sup>.

Gigaphoton investigated the difference in CE and ionization under different PP laser conditions<sup>91,99</sup>. As shown in Fig. 8, a CE of 4.7% was obtained under 10 ps condition. Additionally, in the case of 10 ps PP, the ionization rate is very high even with low CO<sub>2</sub> pulse energy. It can explain the fact that the CE of 10 ps PP is greater than that under the 10 ns PP condition in all CO<sub>2</sub> pulse energy ranges, as shown in Fig. 8a. This optimized high-CE PP technology enabled 20 kW CO<sub>2</sub> laser to realize a 250 W EUV source in several minutes. In 2019, the driver laser system was upgraded by improving the optical design of the pre-amplifier and 3 main amplifiers. As a result, a 27 kW CO<sub>2</sub> laser power was achieved by the upgraded configuration. In the latest data, a higher EUV power greater than 360 W with a CE greater than 5% in short term was demonstrated, and 270 W stable operation in long term was achieved<sup>100</sup>. These basic studies have contributed to the development of the high-power laser system and the design for further EUV power scaling.

Several gas laser systems equipped with CO<sub>2</sub> laser pumping beam have been presented and studied by our team at Wuhan University and Huazhong University of Science and Technology<sup>101–103</sup>. To further improve the output power of the gas laser, a radio-frequency CO<sub>2</sub> laser is used in a MOPA system, which is the same technology as NXE:3100<sup>103</sup>. These works laid the foundation for upcoming research to realize high-power CO<sub>2</sub> lasers. Several types of resonators for the CO<sub>2</sub> laser system were proposed, such as waveguide hybrid resonator<sup>104</sup>, plane-cone resonator<sup>105</sup>, and toric concave mirror resonator<sup>106</sup>. The output beam characteristics of the CO<sub>2</sub> lasers with different resonators were investigated in experiment, respectively. For instance, the output mode of the plane-cone resonator CO<sub>2</sub> laser is approximately a TEM<sub>03</sub> mode. With a resonator length of 3 m and an equivalent aperture radius of 30 mm, the output power of the CO<sub>2</sub> laser can be as high as 2 kW, and it can be used for Sn LPP based on CO<sub>2</sub> laser system to generate EUV light.

As the amplification of CO<sub>2</sub> laser system approaches the technical limit, the output power is difficult to be further increased. As such, a novel

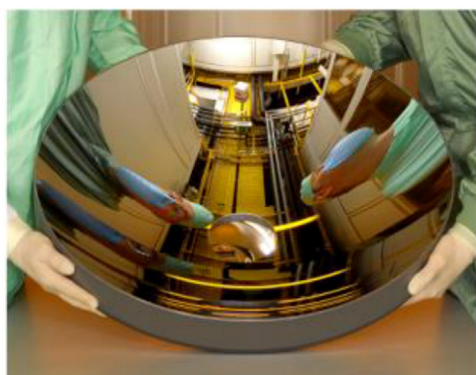


**Fig. 9 | Schematic sketch of multi-fiber coherent synthesis.** Unpublished. © 2022 East China Normal University.

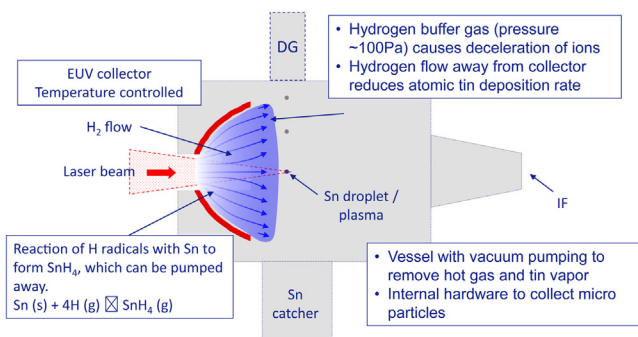
laser system scheme of multi-fiber coherent synthesis has been proposed and is currently under investigation by our team at East China Normal University, as shown in Fig. 9. Based on the currently realized 130 W single-fiber output power<sup>107</sup>, a 27 kW laser power can be achieved by using 207 synthetic Yb-doped fibers. By optimizing the fiber laser amplifier, the number of fibers can be further reduced.

## COLLECTOR MIRROR AND DEBRIS MITIGATION OF Sn LPP SYSTEM

13.5 nm EUV light is absorbed significantly by any materials and environment, even by gasses. Therefore, the whole optical path must be fully reflective in a vacuum environment. Although the plasma is excited by laser over the full  $4\pi$  sr, the typically collector mirror is about  $2\pi$  sr, as shown in Fig. 10<sup>88</sup>. Moreover, in order to select only 13.5 nm EUV light from the plasma emission and increase the reflectivity, the collector mirror (called Bragg mirror) need to be coated with around 50 pairs of Mo/Si multilayers<sup>31</sup>. This selection is realized by giving specific layer thickness  $d$  and reflection angle  $\theta$ , which must satisfy Bragg's law:  $n\lambda = 2d^*\sin\theta$ . The highest experimental reflectivity of Mo/Si multilayer is around 70% at 13.5 nm wavelength<sup>30</sup>. This high reflectivity is mainly attributed to the improvement of multilayer interface quality and a proper barrier layer material.



**Fig. 10 | Picture of a normal incidence EUV collector used by ASML.** Reprinted with permission from ref.<sup>88</sup>. © 2017 Source Workshop.



**Fig. 11 | Illustration of EUV light source vessel with hydrogen gas as buffer.** Reprinted with permission from ref.<sup>88</sup>. © 2017 Source Workshop.

To investigate more possibilities, a variety of multilayer mirrors have been fabricated by our team at Tongji University including Ru/B4C multilayer mirrors<sup>108</sup>, Ru/C multilayer mirrors<sup>109</sup>, Pd/B4C multilayer mirrors<sup>110</sup>, V/Sc multilayer mirrors<sup>111</sup>, Cr/V multilayer mirrors<sup>112</sup>, and Pd/Y multilayer mirrors<sup>113</sup>. All multilayers were fabricated by direct current magnetron sputtering using a linear deposition system<sup>114</sup>. This system can deposit a mirror with 1.2 m length and 0.2 m width at maximum. To study the X-ray performance of those multilayer mirrors, the mirrors were characterized by grazing incidence X-ray reflectometry (GIXR) at different photon energies. For instance, a high X-ray reflectivity of 71% was demonstrated at 10 keV for an Ar-made Pd/B4C multilayer mirror<sup>110</sup>. In a further study, the aging process of the Pd/B4C multilayer mirror was considered<sup>115</sup>. A distinct aging effect was observed on the nitridated samples, after storing the multilayers in air for 6–17 months. Then, a primitive model of upward diffusion of nitrogen and boron was proposed to explain the aging effects of the nitridated structure. The studies on multilayer mirrors for X-ray can provide useful guidance for the development of the EUV or the so-called soft X-ray multilayer mirrors.

The collector is close to the plasma generation area in the vacuum vessel of the EUV source. Moreover, the plasma emits not only EUV light but also high-energy ions, as well as unwanted small Sn particle fuel. All of these debris can cause irreversible damage to the reflective coating and further reduce the reflectivity and lifetime of the collector. Therefore, it is necessary to protect the collector due to the high cost of large, curved hyper-smooth multilayer-coated EUV optical elements. ASML has implemented hydrogen buffer gas with a pressure of about 100 Pa around the collector mirror as shown in Fig. 11<sup>88</sup>. This configuration enabled the buffer gas to decelerate tin ions and reduce tin deposition onto collec-

tor surface by a chemical reaction:  $\text{Sn (s)} + 4\text{H (g)} = \text{SnH}_4 \text{ (g)}$ .  $\text{SnH}_4$  was in vapor state which was removed from the vessel by a vacuum pumping system.

In addition, a capping layer on top of the Mo/Si multilayer is needed to enhance durability and reliability of collector. Ruthenium is one option for such capping layer material due to its relative stability under EUV and hydrogen conditions. Other advanced materials with high EUV transmission and high temperature stability have been investigated<sup>116,117</sup>. Gigaphoton have selected one best-durability capping layer material from three potential candidates in corporation with “New SUBARU” at Hyogo University, Japan. After 600 mins in EUV source, the collector still have 95% reflectivity<sup>116</sup>. Additionally, research team at ASML found that polysilicon and metal-silicide film have good performance in high temperature environment<sup>117</sup>. In 2009, our team at Tongji University started a study on capping layer of Mo/Si multilayer mirrors. The thickness and material of the capping layer were optimized by analyzing the distribution of the standing-wave electric field of the multilayer layers<sup>118</sup>. The calculation results showed that a certain thickness of capping layer always corresponds to an optimal uppermost Si thickness. At a wavelength of 13.36 nm, the theoretical reflectivity of 50-pair Mo/Si multilayer layers is 74.47%.

With the improvements in the collector protection system, longer lifetime has been achieved in actual light sources systems. The collector mirror degradation rates have fallen below 0.1% per giga-pulse (GP) of EUV light<sup>31</sup>. This means that the reflectivity rate decreases by 0.1% for every  $5.6 \times 10^6$  hours of continuous work under a 5 kHz laser pulse. In the latest results, this rate was updated to 0.05% GP with the power at 250 W by ASML<sup>119</sup>. Further improvements to reduce the rate are under development, including a re-design of the plasma to direct tin towards the exhaust and stop tin vapor from reaching collector.

## DROP GENERATOR AND METROLOGY OF Sn LPP SYSTEM

Inside the EUV source chamber, extremely small and stable Sn droplets are continuously released from the drop generator (DG) at a high frequency. As illustrated in Fig. 12, tin raw material with high purity was loaded in a vessel connected to the nozzle by a modulator and a filter<sup>88</sup>. Before the generation of Sn in jet, the raw material was heated above melting point and flowed through the filter prior to the nozzle. An inert gas was applied to provide power source and finish this flow. Then the tin jet was modulated into droplets by mechanical vibration, and further sprayed out of the nozzle. The separation space and the size of droplets could be well-controlled by the generator pressure and vibration frequency.

The main requirement for the droplet generators is that, the delivered droplet targets should exact the same size, and arrive at the same location with the same repetition rate as the laser pulses. In addition, Sn droplets should have sufficient distance between each other, to reduce the effects of plasma generation on the next droplet. In order to achieve this sufficient distance, ASML have applied a special modulation method by coalescing multiple small droplets into larger droplets at larger separation distance. The detailed process of this modulation is demonstrated in Fig. 13<sup>88</sup>. ASML's third-generation droplet generator has been released, and the key performance parameters of the droplet generator are shown in Table 1<sup>20</sup>. It has produced droplets with a diameter of 27  $\mu\text{m}$  and around 2,700 hours of average lifetime.

With third-generation droplet generator configurations, Sn was re-filled with an interruption of the scanner run which increased the downtime. In order to reduce downtime, the new inline refill supply system with two pressurized Sn tanks has been developed by ASML, as shown in Fig. 14<sup>92</sup>. When the Sn in one tank is empty, the system will automatically

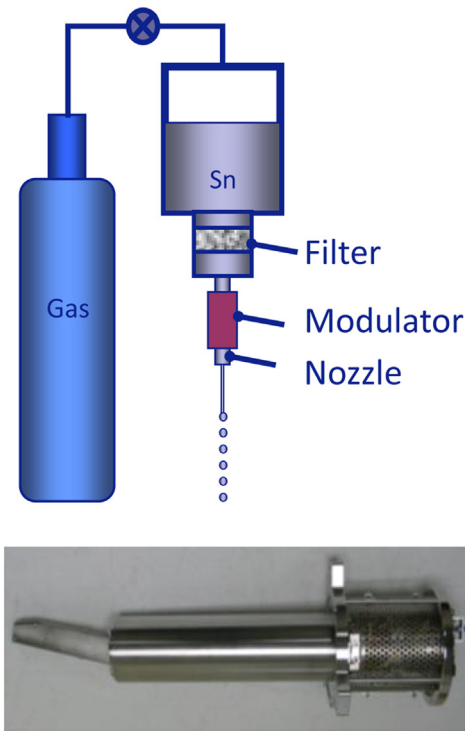


Fig. 12 | Illustration of a Sn droplet generator. Reprinted with permission from ref.<sup>88</sup>. © 2017 Source Workshop.

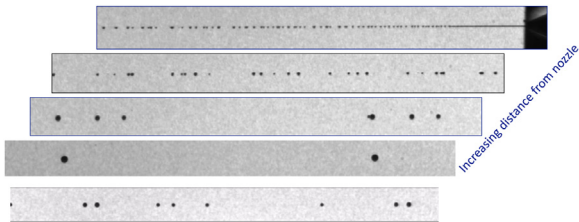


Fig. 13 | ASML’s method of coalescing multiple small droplets into larger droplets. Reprinted with permission from ref.<sup>88</sup>. © 2017 Source Workshop.

Table 1 | Key performance parameters of the droplet generator<sup>120</sup>.

| Performance paraneter | Generation III                               |
|-----------------------|----------------------------------------------|
| Key Features          | Restart & Refill capable                     |
| Run-time              | ~780 hours (average)<br>2700 hours (maximum) |
| Start-up yield        | >95%                                         |
| Availability          | 95%                                          |
| Droplet diameter      | 27 μm                                        |

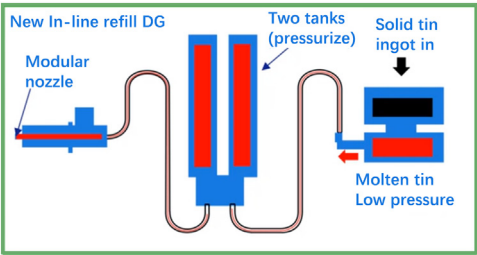


Fig. 14 | Illustration of the new inline refill supply system by ASML. Reprinted with permission from<sup>92</sup>. © 2021 SPIE.

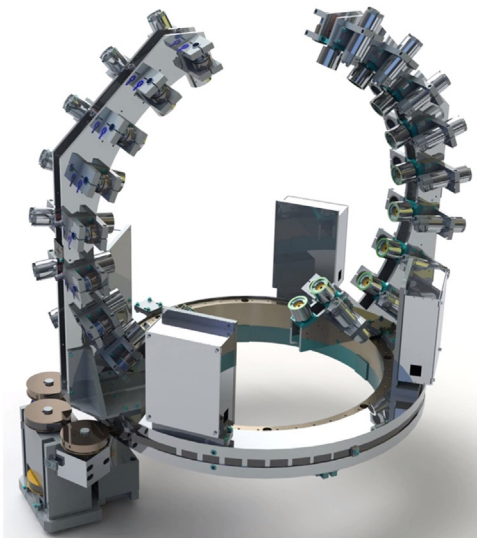


Fig. 15 | Measurement apparatus of ion inputs and EUV emission. Reprinted with permission from ref<sup>121</sup>. ©2020 SPIE.

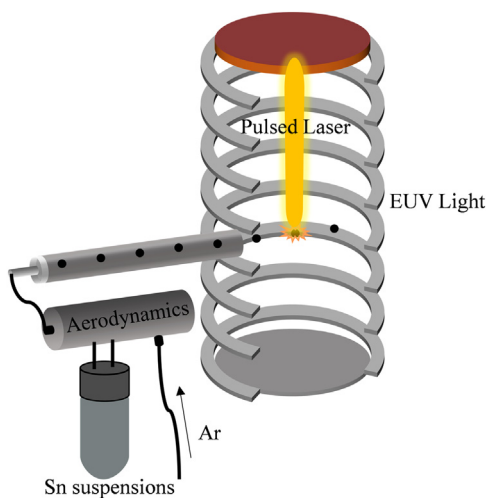
switch to the other tank, and the scanner keeps running. This new configuration can deliver EUV around the clock, without interruption to reload, melt, and re-pressurize the Sn. Furthermore, it makes the Sn droplets constantly available for laser-plasma interaction at 50 kHz repetition rate, and the lifetime is increased to beyond 3,380 hours.

To characterize plasma distribution and reduce debris, a measurement apparatus has been proposed by researchers at ASML US<sup>121</sup>. It consists of multiple sensors installed on a rotating frame, to monitor tin ion inputs and EUV emission, as shown in Fig. 15. Then the data of ion inputs is introduced into a computational fluid dynamics model, for iterating new hardware components or new H<sub>2</sub> flow configurations. Furthermore, this process will be able to optimize the protection of the collector.

SUMMARY AND FUTURE OUTLOOK

EUVL has proven to be the leading technology for the manufacture of next-generation semiconductor devices, and the Sn LPP driven by CO<sub>2</sub> laser system is the best EUV light source so far with high CE and high output power. For laser power scaling, MOPA PP technology has been validated as the best solution, achieving ≥ 40 kW laser power output. MOPA PP is able to provide a stabilized EUV power of 250 W with 6% CE. Using advanced control technologies, stable dose with less than 0.1% dose error has been achieved. Normal-incidence collector mirrors with about 2π sr light collection and high average reflectivity are showing increasingly long lifetimes, with below 0.1% degradation per giga-pulse. Furthermore, enhanced collector protection and in situ cleaning technologies are under





**Fig. 16 | Schematic sketch of our proposed nano-Sn-generator.** Unpublished. © 2022 East China Normal University.

development, both of which are expected to further increase system lifetime. In addition, the stable Sn droplets with diameter of approximately  $27\ \mu\text{m}$  generated by a well controlled droplets generator system, will form a uniform plasma distribution. It further reduces debris introduced onto the collector.

In the latest reports, the main parameters with key influences on the EUV source have been updated. For example, the dose-controlled EUV power reaches 420 W at a high duty cycle and the open loop EUV power of 500 W was demonstrated. Additionally, the collector mirror degradation rate has been updated to below 0.05% GP in ASML's NXE: 3600D. Furthermore, a new Sn refill system enabling Sn reload and transfer inline has been developed, and it is expected to reduce the system downtime. Research and development of the EUV light source, as well as each of its components, are still being pursued even now that ASML has put the scanner on the market. For example, the development of LPP light sources using  $2\ \mu\text{m}$  wavelength lasers is a recent topic<sup>122–124</sup>, CE in excess of 3 % has been achieved from a Sn plasma driven by a  $2\text{-}\mu\text{m}$ -wavelength laser system<sup>122</sup>.

Recently, a new strategy for EUV source has been presented by our team at East China Normal University, which is a combination of a multi-fiber laser coherent synthesis system and a nano-Sn generator. The method of multi-fiber coherent synthesis offers a significant advantage of providing a high peak power, high average power, and high conversion efficiency simultaneously. The technical principle has been verified in previous studies<sup>125</sup>. However, there are two remaining problems to be solved: the coherent addition techniques of multi-fibers, and Sn debris cleaning. It will be the focus of our future studies. For nano-Sn-generator, the direct contact of Sn nanoparticles with laser light allows the Sn to be transformed into multiple excited states more quickly, which is the main source of EUV light<sup>126</sup>. Stable nanoparticle emission can be achieved by adding noble gas aerodynamic systems. Indeed, a critical problem arises in this process: more Sn debris are generated in the light source system, giving greater difficulty to the cleaning work. Upcoming research of our team will focus on the collection of debris. These innovations we proposed have a potential to enhance both the input laser power and Sn target conversion efficiency. In addition, it is a challenge to characterize the size uniformity, space-time coordinate stability, velocity, and expansion kinetics of Sn droplets. Our research team at Wuhan University has proposed a solution, with ongoing work aiming at monitoring the real-time state of droplets using the

ultrafast imaging technology developed by our team. It should be noted that, instead of imaging every droplets, a sampling selection strategy could be employed to improve monitoring efficiency. Yet another direction with high research potential is to create higher plasma density of Sn to realize more Sn-ion energy level transitions. To achieve this end, our team at East China Normal University proposes a nano-Sn-generator, as illustrated in Fig. 16. Diffusing Sn nanoparticles into water in a certain proportion can obtain Sn suspensions. The suspensions are atomized and dried in an Ar atmosphere, then squeezed out to the laser focus by aerodynamics. It allows the laser to interact with a high concentration of Sn nanoparticles. The nano-Sn target can maximize the reaction area with the laser to enhance the efficiency, which can greatly increase the output power of the EUV light.

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**Supplementary materials** Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.chip.2022.100019](https://doi.org/10.1016/j.chip.2022.100019).

**Funding** This study was supported by the National Key R&D Program of China (2019YFB1704600).

**Declaration of Competing Interest** The authors declare no competing interests.

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